AFYAPULSE

Comprehensive Software Requirements Specification (SRS) / Product Requirements Document (PRD)

Early Health Signal Network - MVP / Hackathon Build

Architecture: Next.js + TypeScript + MongoDB Atlas + Mongoose + Auth.js + Groq

Authentication: One Unified Entry for Citizens and Public Health Officers

Document status: Approved working specification. This version supersedes the earlier PostgreSQL/Prisma architecture and the earlier interpretation of separate citizen/officer public signup pages. It is the product-level source of truth before the individual developer work packages.

Version 3.0 • September 2026

1. Document Purpose, Audience & Authority

This document defines what AfyaPulse V1 must do, how its main user experiences behave, the approved technical architecture, privacy and safety boundaries, core data concepts, integration rules, and the acceptance criteria for the hackathon MVP.

Primary audience

- All four AfyaPulse developers.
- Product/design contributors.
- Hackathon presenters and QA testers.
- Future maintainers who need to understand the MVP boundary.

Document authority

- Where an older AfyaPulse document says PostgreSQL or Prisma, this SRS replaces it with MongoDB Atlas + Mongoose.
- Where an older document describes two disconnected public signup pages, this SRS replaces that with one unified public authentication entry.
- Individual developer work packages may add implementation detail but must not contradict this SRS.

2. Product Vision, Problem & Value Proposition

Product: AfyaPulse - Early Health Signal Network.

Tagline: Turn small health signals into early warnings.

Problem

Many health events begin as scattered individual symptoms. A single report may not mean much, but a privacy-protected pattern across time and place can become useful public-health intelligence. Citizens also need a simple way to keep their own health timeline without turning the product into a diagnostic tool.

Solution

AfyaPulse lets citizens record symptoms and selected vitals. With explicit consent, eligible fields contribute to de-identified aggregate signals. Authorized Public Health Officers can monitor unusual aggregate activity, investigate, coordinate response and publish reviewed notices to citizens.

Core product loop

Citizen reports → data stored securely → consent/privacy rules applied → aggregate signals → unusual pattern detected → Health Officer investigates → response → reviewed Public Health Alert → citizen receives alert.

Product principles

- Privacy before surveillance.
- Human review before public-health claims.
- Useful personal tracking even when community contribution is declined.
- Transparent detection over opaque AI decisions.
- One coherent product shared by citizens and officers, with different permissions.

3. MVP Scope & Explicit Non-Goals

In scope

- Citizen web application.
- Public Health Officer portal.
- Unified authentication entry.
- Citizen onboarding and consent.
- Symptom/event reporting and conditional body map.
- Manual vital entry.
- Personal timeline/trends/insights.
- Privacy-safe aggregation.
- Officer dashboard, map, warnings, Explorer, investigations and analytics.
- Response workflow and reviewed public alerts.
- Officer access request and higher-level approval.
- Notifications, reports/export and simulation mode.
- Lightweight Groq-generated summaries.
- Responsive deployment.

Not in V1 / post-demo roadmap

- Full AfyaPulse Platform Admin Portal.
- Wearable-device integrations.
- Continuous GPS tracking.
- Exact citizen home-location mapping.
- SMS-first emergency infrastructure.
- Advanced clinical diagnosis.
- Automated outbreak declaration.
- Large-scale research marketplace.
- Complex machine-learning epidemiology models.
- Full multilingual implementation, clinic referral network and long-term population-scale infrastructure.

4. User Types, Roles & Trust Boundaries

Actor	Purpose	Trust boundary
Citizen	Tracks own health and receives reviewed public alerts.	Can access own private records only; optional community contribution is consent-based.
Public Health Officer	Reviews aggregate signals, investigates and coordinates response.	Must be approved; access limited by capability and jurisdiction.
Higher-level / Approver Officer	Reviews subordinate officer access and selected publication/workflow actions.	Approval authority is configured; not every officer is an approver.
System / Internal Service	Runs aggregation, detection, notifications and integrations.	Server-only, least-privilege operations.
Groq AI Provider	Produces constrained summaries from approved inputs.	Never receives direct citizen identity or becomes source of truth.

Authorization principle

Who you are determines WHAT you can do; jurisdiction determines WHERE you can see.

5. Approved Technical Architecture

Layer	Approved technology
Language	TypeScript
Full-stack framework	Next.js + React
Styling	Tailwind CSS
UI components	shadcn/ui
Icons	Lucide
Database	MongoDB
Managed database	MongoDB Atlas
ODM	Mongoose
Authentication	Auth.js
Validation	Zod
Charts	Recharts
Maps	React Leaflet + Leaflet
Map tiles	OpenStreetMap-compatible tile source for V1/demo
AI	Groq API - server-side only
Deployment	Vercel + MongoDB Atlas
Source control	Git + GitHub

Architecture rules

- One repository and shared application data model.
- Frontend components do not directly connect to MongoDB.
- Protected operations execute through server services/actions/API routes.
- No permanent localStorage/mock path for data that belongs to the shared backend.
- No PostgreSQL/Prisma dependency in the approved V1 architecture.

6. Unified Authentication Experience

AfyaPulse uses one public authentication entry. The goal is one product identity while keeping officer privileges strongly separated from ordinary citizen account creation.

Main auth page

- AfyaPulse logo/branding and concise trust message.
- Citizen Sign In and Sign Up are the primary/default controls.
- Forgot Password / verification entry where implemented.
- Below the main citizen controls: Are you a Public Health Officer?
- Two officer actions: Sign in as Health Officer and Request Health Officer Access.

Citizen authentication flow

Main Auth → Citizen Sign Up / Sign In → Onboarding if needed → Consent → Citizen Dashboard.

Existing Health Officer flow

Main Auth → Sign in as Health Officer → authenticate → server verifies approved OfficerProfile, jurisdiction and capabilities → Officer Overview.

New Health Officer flow

Main Auth → Request Health Officer Access → account/details → organization → position/designation → jurisdiction → verification evidence → review → submit → awaiting higher-level approval → activation → Health Officer sign-in.

Mandatory security behavior

- Citizen signup never contains a role selector capable of granting officer privileges.
- Creating an account is not equivalent to becoming an officer.
- Pending, rejected or incomplete officer requests cannot access protected officer intelligence.
- Officer role, jurisdiction and capabilities are authoritative server-side values.

7. Citizen Landing Page Requirements

The landing page must explain the value of AfyaPulse before account creation and establish privacy expectations.

Required sections

- Hero: Turn Small Health Signals Into Early Warnings.
- Primary CTAs: Get Started and Learn More.
- Problem statement.
- How AfyaPulse Works.
- Personal Health Features.
- Privacy & Trust.
- Public Health Impact.
- Final CTA.
- Footer.

Transparency requirement

Before signup, users should understand that personal records remain private and that, when they consent, privacy-protected/aggregated health signals may contribute to public-health intelligence monitored by authorized officers.

Approved trust concept

Public Health Officers monitor authorized aggregated population signals - not a citizen's private health timeline.

8. Citizen Onboarding & Consent

Onboarding sequence

Personal → Location → Health Profile → Privacy → Complete.

Personal

- Full name.
- Date of birth.
- Sex/gender only where appropriate and optional as designed.

Location

- Country.
- State.
- LGA/district.
- City/community.
- Approximate location may be optional.
- Exact home address is not required for MVP.

Optional Health Profile

- Blood group.
- Allergies.
- Existing conditions.
- Current medications.
- Emergency contact only if product team retains it.

Consent model

- Provide a clear first-time consent experience, not repeated intrusive prompts.
- Personal tracker remains usable when optional community contribution is declined.
- Separate purposes where applicable: community contribution, approximate location, AI analysis and research.
- Remember choices and expose them under Settings → Privacy & Consent.
- Re-consent only when a materially new data use/purpose/version requires it.
- Consent history must be representable in the backend, not a single permanent boolean.

9. Citizen Dashboard Requirements

The dashboard answers: What has been happening with my health recently, and what should I record next?

Required content

- Latest health signal.
- Symptoms this week.
- Latest temperature or other recent vital.
- Personal trend/summary.
- 7-day health activity visualization.
- Recent signals/events.
- Quick actions.
- Primary CTA: + Report Health Signal.

States

- First-time/empty state.
- Loading state.
- Partial data state.
- Error/retry state.
- Normal populated state.
- AI insight unavailable fallback.

10. Report Health Signal - Detailed Functional Requirements

The reporting flow must not assume every symptom belongs to a body part.

Step 1 - What are you experiencing?

- Support multiple symptoms in one HealthEvent.
- General examples: fever, fatigue, chills, weakness, dizziness.
- Digestive: vomiting, diarrhea, nausea, appetite changes.
- Respiratory: cough, difficulty breathing, runny nose.
- Skin: rash, itching, swelling.
- Body-specific/pain: headache, chest pain, abdominal pain, back pain, joint pain.
- Other/search/description.

Step 2 - Body location if applicable

- Body map appears only when location adds useful meaning.
- Chest pain can use body map; fever/vomiting/diarrhea need not.
- Provide a text alternative to body-map interaction.

Step 3 - Severity

- 1-10 scale.
- Human-readable grouping such as mild/moderate/severe without diagnosing.

Step 4 - Timing/details

- When it started.
- Duration.
- Recurrence.
- Possible trigger.
- Notes.

Step 5 - Optional measurement

Allow a relevant optional measurement such as temperature where appropriate.

Step 6 - Review and submit

- Show selected symptoms, body area if any, severity, timing, duration, notes and current privacy/community-contribution status.
- On success, persist to personal timeline.
- Eligible fields may enter privacy-preserving aggregation according to authoritative consent.

11. Manual Vitals Requirements

Vital	Required fields / behavior
Temperature	Value, °C/°F, date/time, optional note/source.
Blood pressure	Systolic + diastolic, mmHg, date/time, optional note/source.
Heart rate	BPM, date/time, optional note/source.
SpO2	Percentage, date/time, optional note/source.
Weight	kg/lb, date/time, optional note/source.

Rules

- Manual entry only for V1.
- Validate plausible input format/range without presenting a diagnosis.
- Not every vital should contribute equally to public-health intelligence.
- Temperature and symptom/time/location patterns may be relevant; weight is primarily personal in V1.
- Wearables are future scope.

12. My Health Requirements

Tabs

Timeline | Trends | Insights

Timeline

- Chronological symptoms, HealthEvents and vitals.
- Filter/search may be added if time permits.
- Private to the citizen.

Trends

- Symptom frequency.
- Severity over time.
- Recurring symptoms.
- Body areas.
- Vital trends.

Insights

- Non-diagnostic observations only.
- Example: You reported headaches 4 times in the last 14 days.
- Avoid disease claims such as You have migraine/malaria.
- AI insight must degrade gracefully to deterministic text or no-AI state.

13. Citizen Health Alerts & Settings

Health Alerts

- Citizens receive reviewed/published notices, not raw anomaly feeds.
- Show affected broad area, update date, issuing/approving authority, guidance, urgency and duration where available.
- Do not expose raw case data, officer-only anomaly scores, investigation notes, identities or exact locations.

Profile & Settings

- Profile.
- Health Profile.
- Privacy & Consent.
- Security.
- Notifications.
- Preferences.
- Help & Support.

Privacy controls

- Community contribution toggle/choice.
- Approximate-location use.
- AI analysis permission if implemented as a distinct consent purpose.
- Research consent only if introduced.
- Data export/download if in scope.
- Account/data deletion subject to agreed retention/legal requirements.

14. Public Health Officer Access Request & Verification

Officer access is not self-service privilege escalation. AfyaPulse can be used by existing public-health personnel without requiring AfyaPulse employees in every location.

Request fields

- Full name.
- Official email.
- Phone.
- Organization.
- Position/designation.
- Requested jurisdiction.
- Official ID/letter or verification evidence metadata/reference.
- Review and submit.

Verification/status experience

- Identity Verification.
- Organization Verification.
- Jurisdiction Confirmation.
- Awaiting Approval.
- Final Activation.
- Approval chain/status.

Approval model

- Local/LGA officer request can be reviewed by an authorized higher-level/state supervisor.
- State-level requests can require higher-level review.
- Optional regional/zonal level can be configured.
- Federal/national onboarding can use institutional verification.
- The hierarchy must be configurable rather than hard-coded to one organizational assumption.

15. Officer Portal Navigation & Application Shell

Desktop sidebar

- Overview.
- Signal Map.
- Early Warnings.
- Signal Explorer.
- Investigations.
- Analytics.
- Response.
- Public Alerts.
- Reports.
- Notifications.
- Approvals - role dependent.
- Simulation.
- Profile & Settings fixed at bottom.

Mobile navigation

Home | Map | Alerts | Reports | More

Application-shell requirements

- Consistent active navigation state.
- Officer identity, organization/jurisdiction and verified status visible where useful.
- Responsive sidebar/drawer.
- Unread notification count.
- Permission-hidden items must still be protected server-side.

16. Officer Overview Dashboard

The Overview answers: What unusual health activity is happening in my jurisdiction, and what needs my attention?

Required elements

- Active Signals metric.
- Requires Review metric.
- New Reports metric.
- People Reached or other approved aggregate impact metric if data supports it.
- Health Activity Map preview.
- Top Symptoms over selected period.
- Recent Alerts.
- What Changed Today? summary.
- Timeframe selector.
- Officer greeting and jurisdiction context.

Data rule

Every metric must come from privacy-safe, jurisdiction-authorized aggregate data. The dashboard must not fetch raw citizen records and aggregate them in the browser.

17. Live Signal Map

Technology

React Leaflet + Leaflet + OpenStreetMap-compatible tiles. No Google Maps or Mapbox API is required for the hackathon MVP.

Map behavior

- Show aggregate health activity by area/heat/chloropleth-style visualization as appropriate.
- Country/State/LGA/community drill-down only as authorized.
- No individual citizen pins.
- No exact home coordinates.
- No continuous citizen GPS tracking.

Filters

- Location.
- Symptom.
- Health condition/context.
- Age group.
- Sex where permitted/safe.
- Time range.
- Severity.
- Report type.

Privacy

- Small/unsafe groups are suppressed.
- Map and Signal Explorer share backend authorization/privacy logic.
- Provide a non-map textual summary of essential information.
- Map owner in implementation: Developer 2; location collection: Dev 1; aggregation: Dev 3; privacy/jurisdiction enforcement: Dev 4.

18. Early Warning Center & Detection Semantics

Example signals

- Fever + cough increase.
- Diarrhea increase.
- Rash cluster.
- Respiratory symptom rise.
- A permitted health-context + symptom combination.

Statuses

New | Monitoring | Requires Review | Escalated | Closed

Language requirements

- Use Pattern noticed, activity above recent baseline, requires review, monitoring.
- Do not label an algorithmic anomaly as a confirmed outbreak/disease.
- Professional review is required before public communication.

Detection concept

Eligible aggregate → current count/rate → recent baseline → deterministic comparison/rule → warning candidate.

19. Signal Explorer

Purpose

Allow authorized officers to answer cross-signal questions without exposing citizen-level records.

Filters

- Location.
- Health condition/context.
- Multiple symptoms/combinations.
- Age bands/range.
- Sex where permitted.
- Time.
- Severity.
- Recurring/new.
- Body area.
- Duration.
- Selected vitals.
- Comparison period/area.

Example

People aged 40-60 + recorded diabetes context + headache + Ife Central + last 7 days - represented only as a privacy-safe aggregate query.

Results

- Matching aggregate signal/count/rate where safe.
- Change vs baseline.
- Common related symptom.
- Age distribution.
- Peak period.
- Severity distribution.
- Map/trend.
- Related symptoms.

Privacy response

If filters create a group below the configured privacy threshold, return Insufficient aggregated data rather than the hidden exact count.

Optional productivity features

- Save Query.
- Compare.
- Watch/alert when pattern rises significantly above baseline.

20. Investigations, Analytics & Response

Investigations list

- Signal.
- Location.
- Status.
- Assigned officer/team.
- Last updated.

Investigation detail

- Overview.
- Trends.
- Map.
- Related Symptoms.
- Demographics.
- Notes.
- Actions.
- Time window.
- Privacy-safe aggregate report total/rate.
- Change from baseline.
- Confidence/evidence description.
- Assign, note, escalate, monitor, close.

Analytics

- Symptoms.
- Conditions/health context.
- Demographics.
- Trends.
- Geographic Comparison.
- Vitals.

Response Management

Detected → Reviewing → Monitoring → Escalated → Response → Resolved. Actions must be persisted and auditable.

21. Public Health Alerts, Reports, Notifications & Approvals

Public Health Alerts

- Draft.
- Submit/review/approve according to capability.
- Publish.
- Update.
- Expire/close.
- Affected broad area.
- Guidance.
- Urgency.
- Duration.
- Published notices become Citizen Health Alerts.

Reports & Export

- Daily Summary.
- LGA/Regional Report.
- Symptom Trends.
- Investigation Report.
- Response Report.
- Custom/scheduled later if time permits.
- All exports obey jurisdiction and privacy rules.

Notifications

- Access Request - Action Required.
- New Early Warning.
- Investigation Assigned.
- Alert Approval Request.
- System/security update.
- Unread/read state and View All.

Approvals

- Pending / Approved / Rejected.
- Applicant, requested role, jurisdiction, submitted time, status and Review.
- Approve / Reject / Request More Information.
- Every decision is permission-checked and auditable.

22. Simulation / Hackathon Demonstration Mode

Purpose

The hackathon must demonstrate the full system even if real-world signal volume is unavailable.

Demo sequence

Inject Test Data → Watch Aggregation/Detection → Review Early Warning → Investigate → Record Response → Generate Report → Publish reviewed demo alert.

Rules

- Clearly label Simulation Mode / Demo Data / Not real patient information.
- Seed data must be synthetic.
- Simulation cannot be confused with real public-health records.
- Developer 3 supplies deterministic seed/reset data; Developer 4 protects/integrates; Developer 2 exposes the demo controls.

23. MongoDB / Mongoose Data Architecture

MongoDB modeling approach

Use embedding when data is bounded, owned by one aggregate and normally read together. Use references for shared, independently growing, reusable or cross-linked entities. Avoid blindly recreating relational join tables as separate collections.

Core models/collections

- User.
- CitizenProfile.
- HealthProfile.
- ConsentRecord.
- Symptom.
- HealthEvent.
- VitalReading.
- Organization.
- Jurisdiction.
- OfficerProfile.
- AccessRequest.
- ApprovalDecision.
- AggregatedSignal or aggregate snapshot if useful.
- EarlyWarning.
- Investigation.
- ResponseAction.
- PublicAlert.
- Notification.
- AuditLog.

Critical schema rules

- One HealthEvent supports multiple symptoms.
- Consent retains purpose/version/state/history.
- Jurisdiction supports parent-child hierarchy.
- Officer privilege is not stored as an unverified client-controlled field.
- Simulation records are marked.
- PublicAlert supports draft/review/published/expired lifecycle.
- Indexes must reflect actual citizen-history and aggregate query patterns.

24. Backend Services & Data Contracts

Suggested service modules

- auth.
- citizen-profile.
- health-profile.
- consent.
- health-events.
- symptoms.
- vitals.
- jurisdictions.
- officer-access.
- signals.
- warnings.
- explorer.
- investigations.
- responses.
- public-alerts.
- reports.
- notifications.
- audit.

Every shared service contract documents

- Purpose.
- Authentication requirement.
- Role/capability.
- Input schema.
- Output schema.
- Validation.
- Errors.
- Pagination/filter behavior.
- Privacy behavior.
- TypeScript/Zod type where practical.

Server identity rule

Private citizen operations derive the current citizen from the authenticated server session. A browser-provided userId must not be trusted as proof of ownership.

25. Aggregation & MongoDB Query Requirements

Target pipeline

HealthEvent → authoritative consent eligibility → approved fields → de-identification → time/geography bucket → privacy threshold → MongoDB aggregation → baseline/detection → officer-safe result.

Requirements

- Use MongoDB aggregation pipelines/server services for population grouping.
- Do not download all citizen events to officer frontend and aggregate there.
- Support time, geography, symptom, severity, permitted demographic/context and selected vital dimensions.
- Add compound indexes based on measured query patterns.
- Paginate growing lists.
- Avoid unbounded embedded arrays and oversized documents.
- Persist/materialize aggregate snapshots only when useful; document refresh behavior.

26. Privacy, Consent & Security Requirements

Privacy requirements

- No citizen name, email, phone, account ID or exact home coordinate in officer aggregate responses.
- No generic officer get-all-citizens surveillance endpoint.
- Minimum aggregate group-size threshold is configurable.
- Narrow queries cannot reveal the suppressed count.
- Reports/export obey the same privacy rules as screens.
- Logs must not contain passwords, tokens, database URI, Groq key or unnecessary health payloads.

Consent requirements

- Community contribution is optional.
- Server checks authoritative ConsentRecord rather than trusting a request flag.
- Different purposes remain separable where implemented.
- Changes are handled according to agreed future-processing/retention rules.

Security requirements

- Auth.js sessions/cookies configured safely.
- Server-side RBAC/capability checks.
- Officer approval gating.
- Jurisdiction authorization on every protected data path.
- Safe error messages without sensitive existence leakage.
- Secrets remain server-side.

27. Groq AI Architecture & Safety

Architecture

MongoDB aggregate data → deterministic TypeScript calculations → privacy check → Groq summary → UI.

Allowed uses

- Summarize already-computed aggregate changes.
- Generate concise non-diagnostic citizen trend wording.
- Generate officer-readable pattern summaries from approved aggregate data.

Disallowed uses

- Diagnosing a citizen.
- Sending direct identity/private raw timelines.
- Inventing counts/percentages/baselines.
- Deciding warning status.
- Automatically declaring an outbreak.
- Blocking the core product when AI is unavailable.

Fallback

When Groq is unavailable, return deterministic text based on the computed metrics. Reporting, aggregation, warnings, investigations and public alerts must continue to work.

28. Design System & UI Standards

Token	Approved value / usage
Deep Navy	#08214A - navigation, headings, trust.
Primary Blue	#1267E8 - primary actions/active states.
Cyan/Teal	#20BFD0 - health-tech accent.
Off White	#F4F8FD - page surfaces.
White	#FFFFFF - cards/content.
Green	#22A06B - normal/success.
Amber	#D99000 - warning/review.
Red	#C63C3C - urgent/critical/error only.
Typography	Inter; fallback system-ui, -apple-system, Segoe UI, sans-serif.
Cards	12-16px rounded, subtle border/shadow.
Icons	Lucide-style consistent iconography.

Typography

- 600-700 for titles/important metrics.
- 400-500 for body/interface text.
- Desktop body generally 14-16px minimum in implementation.
- Sentence case preferred.

29. Responsive & Accessibility Requirements

Viewport	Expected behavior
320-374px	Small mobile; no page-level horizontal overflow; forms/actions fit.
375-480px	Primary mobile experience; citizen bottom nav and officer compact navigation.
768-1023px	Tablet; adaptive grids and collapsible officer navigation.
1024-1439px	Laptop; full functional layouts.
1440px+	Desktop; approved sidebar/dashboard direction.
Large desktop	Use additional space for maps/analytics while preserving readable content widths.

Accessibility

- Semantic headings/landmarks.
- Labels for form controls.
- Keyboard-accessible core flows.
- Visible focus states.
- Sufficient contrast.
- Do not rely on color alone for status.
- Reduced-motion support where animation exists.
- Body-map text alternative.
- Chart labels/text summary.
- Map textual essential summary.

30. Required Application States & Error Handling

Every major screen/flow must intentionally support

- Loading.
- Empty/no data.
- Success.
- Validation error.
- Server/network error.
- Unauthorized/forbidden.
- Expired session.
- Privacy-suppressed aggregate.
- No matching results.
- AI unavailable.
- Pending officer approval.
- Rejected/request-more-information officer status.

Error principles

- Do not expose stack traces or database/provider secrets to the browser.
- Preserve user-entered form data on recoverable errors.
- Use actionable retry/re-authentication messaging.
- A privacy-denied state must not reveal the hidden sensitive value.

31. Four-Developer Responsibility Matrix

Developer	Primary ownership	Critical collaboration
Developer 1 - Citizen App Lead	Landing, unified public auth UI/citizen path, onboarding/consent UI, dashboard, reporting, vitals, My Health, citizen alerts, profile/settings.	Consumes Dev 3 services; Dev 4 auth/privacy/AI; routes officer actions to Dev 2 flow.
Developer 2 - Officer Portal Lead	Officer access UI, verification status, dashboard, map, warnings, Explorer, investigations, analytics, response, alerts, reports, notifications, approvals, simulation, officer settings.	Consumes Dev 3 aggregates/workflows and Dev 4 authorization/privacy; publishes alerts consumed by Dev 1.
Developer 3 - Backend & Database Lead	MongoDB Atlas, Mongoose models/indexes, server data services, aggregate query foundation, workflows, notifications/audit, seed/demo data.	Supplies data contracts to Dev 1/2; authoritative models to Dev 4.
Developer 4 - Auth, Security, AI & Integration Lead	Auth.js, RBAC/capabilities, approval gating, jurisdiction, consent/privacy enforcement, deterministic detection, Groq, integration tests, deployment.	Secures Dev 3 services, supplies safe data/AI to Dev 1/2, owns final integration.

32. Environment, Secrets & Deployment

Expected environment variables

```
MONGODB_URI=<secret>
AUTH_SECRET=<secret>
GROQ_API_KEY=<secret>
RESEND_API_KEY=<optional-secret>
```

Rules

- Real values never committed to GitHub.
- Secrets never use NEXT_PUBLIC_.
- .env.example contains placeholders only.
- Developer 3 owns database setup; Developer 4 owns production security/integration.
- Deploy Next.js to Vercel and connect MongoDB Atlas.
- Run post-deployment smoke tests for citizen and officer core flows.

33. End-to-End Acceptance Scenarios

Scenario A - Citizen

- Open unified auth.
- Create citizen account.
- Complete onboarding and consent.
- Report fever + headache + fatigue in one event.
- Add temperature.
- See event in My Health.

Scenario B - Community signal

- Eligible synthetic events are aggregated.
- Privacy threshold is satisfied.
- Current activity exceeds configured baseline rule.
- Early Warning is created/updated with safe wording.

Scenario C - Officer

- Approved officer uses Sign in as Health Officer.
- Officer sees only authorized jurisdiction.
- Dashboard/map/Explorer show privacy-safe aggregates.
- Officer opens investigation and records response.

Scenario D - Public alert

- Authorized officer drafts/reviews/publishes notice.
- Published alert appears to affected citizen audience.
- Draft/internal notes remain hidden.

Scenario E - Officer onboarding

- Applicant selects Request Health Officer Access.
- Submits organization/designation/jurisdiction/evidence.
- Authorized higher officer receives notification.
- Approver approves/rejects/requests more information.
- Decision is auditable; access changes only after authorized approval.

34. Quality Assurance & Security Test Matrix

Functional

- Multi-symptom HealthEvent.
- Conditional body map.
- Vitals.
- Consent history.
- Published alert filtering.
- Officer workflows.
- Notifications/approvals.
- Simulation.

Authorization

- Citizen cannot access another citizen record.
- Citizen cannot call officer endpoints.
- Pending officer cannot access portal intelligence.
- Officer cannot change URL/payload to another jurisdiction.
- Non-approver cannot approve.
- Non-publisher cannot publish.

Privacy

- Officer responses contain no direct identity.
- Map contains no exact citizen coordinates.
- Small groups are suppressed.
- Exports cannot bypass suppression.
- Consent-declined optional community data is excluded.

AI

- Groq key absent from client bundle.
- Only aggregate/de-identified input is sent.
- AI cannot change authoritative numbers/status.
- Provider failure uses fallback.

Responsive/accessibility

- Required viewport matrix passes.
- No major clipping/overflow.
- Keyboard/focus/labels/contrast pass.
- Map/chart/body-map alternatives exist.

35. Final Definition of Done for AfyaPulse V1

AfyaPulse V1 is complete for the hackathon when all of the following are true:

- Unified public authentication entry is implemented.
- Citizen and Health Officer paths are secure and coherent.
- MongoDB Atlas + Mongoose is the only approved application database layer.
- Citizen onboarding/consent/reporting/vitals/My Health/alerts work with real backend data.
- Officer access approval and verification gating work.
- Officer Overview, Map, Warnings, Explorer, Investigations, Analytics, Response, Public Alerts, Reports, Notifications, Approvals, Simulation and Settings are functional at agreed MVP depth.
- Officer intelligence is aggregate, jurisdiction-restricted and privacy-thresholded.
- Deterministic detection creates reviewable warnings.
- Groq improves summaries but is optional to core operation.
- Published reviewed alerts reach citizens.
- Synthetic demo data can demonstrate the full loop.
- Responsive/accessibility requirements are met.
- Secrets are protected.
- The deployed Vercel + MongoDB Atlas application passes the core end-to-end demo.

Final demo story

Citizen report → MongoDB → consent/privacy → aggregate → warning → officer investigation → response → reviewed public alert → citizen receives alert.